

COIMBATORE – 641 105

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

CCW331
BUSINESS ANALYTICS LABORATORY

Name :

Reg. No. :

Branch :

COIMBATORE-641 105

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

*Certified that this is the bonafide record of work done
by..... in the **CCW331 – BUSINESS ANALYTICS**
LABORATOR of..... this institution, as prescribed by
..... Anna University, Chennai for the
..... year/semester during the year 2024- 2025*

Staff-in-charge

Head of the Department

REGISTER No:

Submitted for the Year/Semester Examination of the
institution conducted on

Internal Examiner

External Examiner

COIMBATORE – 641 105

INSTITUTION VISION & MISSION

VISION

To impart quality technical education emphasizing innovations and research with social and ethical values.

MISSION

1. Establishing state-of-the-art infrastructure, effective procedures for recruitment of competent faculty and innovative teaching practices.
2. Creating a conducive environment for nurturing innovative ideas and encouraging research skills.
3. Inculcating social and ethical values through co-curricular and extra-curricular activities.

DEPARTMENT VISION & MISSION

VISION

To become centre of excellence in technical education to produce competent professionals to the dynamic needs in the field of Artificial Intelligence and Data Science.

MISSION

1. Educating the fundamentals of AI&DS by continuously improving the teaching learning methodologies using modern tools.
2. Creating a conducive environment for learning and research-based activities.
3. Inculcating the interdisciplinary skill set required to fulfil the societal need.

CCW331

BUSINESS ANALYTICS

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COURSE OBJECTIVES:

- To understand the Analytics Life Cycle.
- To comprehend the process of acquiring Business Intelligence
- To understand various types of analytics for Business Forecasting
- To model the supply chain management for Analytics.

LIST OF EXPERIMENTS:

1. Explore the features of Ms-Excel.
2. (i) Get the input from user and perform numerical operations (MAX, MIN, AVG, SUM, SQRT, ROUND)
2. (ii) Perform data import/export operations for different file formats.
3. Perform statistical operations - Mean, Median, Mode and Standard deviation, Variance, Skewness, Kurtosis
4. Perform Z-test, T-test & ANOVA
5. Perform data pre-processing operations i) Handling Missing data ii) Normalization
6. Perform dimensionality reduction operation using PCA, KPCA & SVD
7. Perform bivariate and multivariate analysis on the dataset.
8. Apply and explore various plotting functions on the data set.
9. Explore the features of Power BI Desktop
10. Prepare & Load data
11. Develop the data model
12. Perform DAX calculations
13. Design a report
14. Create a dashboard and perform data analysis
15. Presentation of a case study
16. Generate Comparative Statistics (Paired t-test) in Excel.

COURSE OUTCOMES:

- CO1: Explain the real world business problems and model with analytical solutions.
CO2: Identify the business processes for extracting Business Intelligence
CO3: Apply predictive analytics for business fore-casting
CO4: Apply analytics for supply chain and logistics management

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